

Comparing Categorical Frequencies in a Two-Way Table

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|--------|-------|-------|-------|
| 1. 30 | 4. 85 | 7. 16 | 10. — |
| 2. 200 | 5. 30 | 8. 30 | |
| 3. 60 | 6. 40 | 9. 60 | |
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Curriculum

Level: Grade 8 / Pre-Algebra

8.SP.A.1 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

3. If they got 15: divided by the whole survey of 200 students instead of by the 50 students in the sport row
6. If they got 60: used the sport row total of 50 as the base instead of the column total of 75
7. If they got 40: used the sport row total of 50 as the base instead of the column total of 125
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	8 or more hours	Under 8 hours	Row total
Plays a school sport	30	20	50
Does not play a sport	45	105	150
Column total	75	125	200

1. Read one cell.

Go across the row labelled “Plays a school sport” and down the column labelled “8 or more hours.” The cell where they meet holds the students who are in both categories at once.

30 students

That cell is the one every later problem builds on: it is counted once in the row total 50 and once in the column total 75, which is why the two relative frequencies in problems 3 and 6 come out differently.

2. Grand total from the four inner cells.

The four inner cells are disjoint — every surveyed student lands in exactly one of them — so adding all four must reproduce the grand total: $30 + 20 + 45 + 105$. 200 students

Two other routes give the same number and are worth checking with the student: the row totals $50 + 150$, and the column totals $75 + 125$.

3. Row relative frequency: sport players with 8 or more hours.

The question asks about the sport players only, so the base is that row's total, 50 — not the whole survey.

$$\frac{30}{50} = 0.60 \implies 100 \times \frac{30}{50} = 60$$

So 60 percent of the students who play a school sport get 8 or more hours of sleep. 60

Common wrong turn: dividing by 200 gives 15, which answers a different question (what percent of *all* students both play a sport and sleep well).

4. Difference between two cells.

Both counts sit in the “Under 8 hours” column: 105 non-players and 20 players. Subtract the smaller from the larger: $105 - 20$. 85 students

This is a raw-count comparison, and problem 10 asks why a comparison like this one can mislead: the two rows do not have the same number of students in them.

5. Row relative frequency: non-players with 8 or more hours.

Same move as problem 3, but on the second row, whose total is 150.

$$100 \times \frac{45}{150} = 100 \times 0.30 = 30$$

So 30 percent of the students who do not play a school sport get 8 or more hours of sleep.

30

Notice the reversal: the non-players have the *larger* raw count (45 against 30) and the *smaller* percentage.

6. Column relative frequency: well-rested students who play a sport.

This question starts from the sleep variable, so the base is the column total for “8 or more hours,” which is 75.

$$100 \times \frac{30}{75} = 100 \times 0.40 = 40$$

So 40 percent of the students who get 8 or more hours of sleep play a school sport, and the total divided by is the column total 75. 40

Common wrong turn: using the row total 50 gives 60, which is the answer to problem 3 — the same cell, the wrong base.

7. Column relative frequency: short-sleep students who play a sport.

The base is the other column total, 125.

$$100 \times \frac{20}{125} = 100 \times 0.16 = 16$$

So 16 percent of the students who get under 8 hours of sleep play a school sport, and the total divided by is the column total 125. 16

Compare with problem 6: among well-rested students 40 percent are athletes, among short-sleep students only 16 percent are. Reading the table by columns tells the same story as reading it by rows.

8. Gap between the two row relative frequencies.

Problem 3 gave 60 percent and problem 5 gave 30 percent, both measured on their own row. Subtract: $60 - 30$. 30

The units matter: this is 30 *percentage points*, not 30 percent. A gap this large on a survey of 200 students is strong evidence of an association.

9. Median of five class percentages.

Order the five values first: 45, 55, 60, 60, 70. With five values the median is the third one. 60

The class in the table reported 60 percent (problem 3), which sits exactly at the median of the five other classes — so this class is typical, not unusual.

10. Is there an association? (*Open response — flagged for manual review.*)

Model answer. Yes. Among the students who play a school sport, 60 percent get 8 or more hours of sleep; among the students who do not, only 30 percent do. Those two row relative frequencies are far apart, so knowing whether a student plays a sport changes what you would predict about their sleep. That is what an association means.

The raw counts 30 and 45 point the other way and are misleading, because the two rows are not the same size. There are only 50 sport players but 150 non-players, so the non-players have more students in *every* cell simply because there are three times as many of them. Dividing each count by its own row total puts both groups on the same footing — out of 100 — and only then can the two groups be compared fairly.

Grading note: full credit needs (i) the answer yes, (ii) both row percentages quoted, and (iii) an explanation that names the unequal row totals as the reason raw counts mislead. A student who says only “60 is bigger than 30” has compared the percentages correctly but has not explained why the counts fail; give partial credit and ask the follow-up question aloud.